

```
#installation
!pip install pandas
```

```
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: numpy>=1.26.0 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.0.2)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2026.1)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->panda
```

```
import pandas as pd
import numpy as np
```

```
print(pd.__version__)
```

```
2.2.2
```

```
#from list using series
s=pd.Series([1,2,3,4,5])
print(s)
```

```
0    1
1    2
2    3
3    4
4    5
dtype: int64
```

```
#series with indexing
s=pd.Series([1,2,3,4,5],index=["a","b","c","d","e"])
print(s)
```

```
a    1
b    2
c    3
d    4
e    5
dtype: int64
```

```
#dictionary using series
student={
    "name":"khasim",
    "branch":"aiml"
}
s=pd.Series(student)
print(s)
```

```
name      khasim
branch    aiml
dtype: object
```

```
#arithmetic operation
n=pd.Series([10,20,30,40,50])
n+2
n-2
n*2
```

	0
0	20
1	40
2	60
3	80
4	100

dtype: int64

```
#
n=pd.Series([10,20,30,40,50])
n.mean()
```

np.float64(30.0)

n.sum()

np.int64(150)

n.max()

50

n.describe()

	0
count	5.000000
mean	30.000000
std	15.811388
min	10.000000
25%	20.000000
50%	30.000000
75%	40.000000
max	50.000000

dtype: float64

```
#boolean indexing
n[n>20]
```

	0
2	30
3	40
4	50

dtype: int64

n[n<40]

```

      0
0  10
1  20
2  30

dtype: int64
```

```
#vectorized string operations
names=pd.Series(['Khasim','farhan','Jabir'])
names.str.lower()
```

```

      0
0  khasim
1  farhan
2   jabir

dtype: object
```

```
names.str.upper()
```

```

      0
0  KHASIM
1  FARHAN
2   JABIR

dtype: object
```

```
names.str.len()
```

```

      0
0    6
1    6
2    5

dtype: int64
```

```
names.str.startswith('f
')
```

```

      0
0  False
1   True
2  False

dtype: bool
```

```
#dictionary with DataFrame
student={
    "names":["khasim","farhan","jabir"],
    "branch":["aiml","aiml","aiml"],
}
s=pd.DataFrame(student)
print(s)
```

```

      names branch
0  khasim   aiml
```

```
1 farhan  aiml
2 jabir  aiml
```

```
std=[
    {"name":"khasim","branch":"aiml"},
    {"name":"farhan","branch":"aiml"},
    {"name":"jabir","branch":"aiml"}
]
s=pd.DataFrame(std)
print(s)
```

```
   name branch
0  khasim  aiml
1  farhan  aiml
2  jabir   aiml
```

```
arr=np.random.rand(3,4)
c=pd.DataFrame(arr,columns=["a","b","c","d"],index=["x","y","z"])
print(c)
```

```
      a      b      c      d
x  0.698351  0.676268  0.070595  0.576831
y  0.936995  0.265368  0.849132  0.009342
z  0.313745  0.175724  0.946769  0.107951
```

```
#exploring Data Frame
data={
    'name':['khasim','farhan','jabir'],
    'branch':['aiml','aiml','aiml'],
    'age':[20,21,22],
    'score':[90,91,92],
    'grade':['A','B','C']
}
d=pd.DataFrame(data)
print(d)
```

```
   name branch  age  score grade
0  khasim  aiml   20    90     A
1  farhan  aiml   21    91     B
2  jabir   aiml   22    92     C
```

```
d.index
```

```
RangeIndex(start=0, stop=3, step=1)
```

```
d.dtypes
```

```

      0
name  object
branch object
age    int64
score  int64
grade  object

dtype: object
```

```
d.nunique()
```

```

      0
name    3
branch  1
age     3
score   3
grade   3

dtype: int64
```

```
#selecting column
#single column
d['name']
```

```

  name
0  khasim
1  farhan
2   jabir

dtype: object
```

```
d.name
```

```

  name
0  khasim
1  farhan
2   jabir

dtype: object
```

```
#multiple columns
d[['name','score']]
```

```

   name  score  age
0  khasim     90   20
1  farhan     91   21
2   jabir     92   22
```

```
#loc
d.loc[0]
```

```

      0
name  khasim
branch aiml
age     20
score   90
grade   A

dtype: object
```

```
d.iloc[0]
```

0	
name	khasim
branch	aiml
age	20
score	90
grade	A

dtype: object

d.loc[0:2]

	name	branch	age	score	grade
0	khasim	aiml	20	90	A
1	farhan	aiml	21	91	B
2	jabir	aiml	22	92	C

d.loc[0:2, 'name': 'score']

	name	branch	age	score
0	khasim	aiml	20	90
1	farhan	aiml	21	91
2	jabir	aiml	22	92

d.iloc[0]

0	
name	khasim
branch	aiml
age	20
score	90
grade	A

dtype: object

d.iloc[0:2]

	name	branch	age	score	grade
0	khasim	aiml	20	90	A
1	farhan	aiml	21	91	B

d.iloc[0:2, 0:2]

	name	branch
0	khasim	aiml
1	farhan	aiml

d.iloc[-1]

```

      2
name    jabir
branch  aiml
age      22
score    92
grade    C

dtype: object

```

```

#single value
d.iloc[0, 0]

```

```
'khasim'
```

```

import pandas as pd

data={
    'name':['khasim','farhan','jabir'],
    'branch':['aiml','aiml','aiml'],
    'age':[20,21,22],
    'score':[90,91,92],
    'grade':['A','B','C']
}
d=pd.DataFrame(data)

result={
    True:"pass",
    False:"fail"
}
pas={'A':90,'B':80,'D':70}

d['pass']=d['grade'].map(pas)
d['result']=d['grade'].map(result)
print(d)

```

	name	branch	age	score	grade	pass	result
0	khasim	aiml	20	90	A	90.0	NaN
1	farhan	aiml	21	91	B	80.0	NaN
2	jabir	aiml	22	92	C	NaN	NaN

```

d.drop('grade',axis=1,inplace=True)
print(d)

```

	name	branch	age	score	pass	result
0	khasim	aiml	20	90	90.0	NaN
1	farhan	aiml	21	91	80.0	NaN
2	jabir	aiml	22	92	NaN	NaN

```

d.rename(columns={'score':'marks','name':'student'},inplace=True)
print(d)

```

	student	branch	age	marks	pass	result
0	khasim	aiml	20	90	90.0	NaN
1	farhan	aiml	21	91	80.0	NaN
2	jabir	aiml	22	92	NaN	NaN

```
d.isnull()
```

	student	branch	age	marks	pass	result
0	False	False	False	False	False	True
1	False	False	False	False	False	True
2	False	False	False	False	True	True

```
d.notnull()
```

	student	branch	age	marks	pass	result
0	True	True	True	True	True	False
1	True	True	True	True	True	False
2	True	True	True	True	False	False

```
d.isnull().sum()
```

	0
student	0
branch	0
age	0
marks	0
pass	1
result	3

```
dtype: int64
```

```
import pandas as pd
```

```
data={
    'name':['khasim','farhan','jabir'],
    'branch':['aiml','aiml','aiml'],
    'age':['np.nan',21,22],
    'score':[90,91,92],
    'grade':['A','B','C']
}
d=pd.DataFrame(data)
```

```
print(d)
```

	name	branch	age	score	grade
0	khasim	aiml	np.nan	90	A
1	farhan	aiml	21	91	B
2	jabir	aiml	22	92	C

```
d['age'].fillna('not mentioned',inplace=True)
print(d)
```

	name	branch	age	score	grade
0	khasim	aiml	np.nan	90	A
1	farhan	aiml	21	91	B
2	jabir	aiml	22	92	C

```
d.replace('A'np.nan)
```

```
d.sort_values('score',ascending=True)
```

	name	branch	age	score	grade
0	khasim	aiml	np.nan	90	A
1	farhan	aiml	21	91	B
2	jabir	aiml	22	92	C

```
d.sort_values('score',ascending=False)
```

	name	branch	age	score	grade
2	jabir	aiml	22	92	C
1	farhan	aiml	21	91	B

```
d.sort_index(ascending=True)
```

	name	branch	age	score	grade
0	khasim	aiml	np.nan	90	A
1	farhan	aiml	21	91	B
2	jabir	aiml	22	92	C

```
d  
.nlargest(3, "score")
```

	name	branch	age	score	grade
2	jabir	aiml	22	92	C
1	farhan	aiml	21	91	B
0	khasim	aiml	np.nan	90	A

```
employees = pd.DataFrame({  
    'emp_id': [1, 2, 3, 4],  
    'name': ['Alice', 'Bob', 'Charlie', 'Diana'],  
    'dept_id': [10, 20, 10, 30]  
})  
  
departments = pd.DataFrame({  
    'dept_id': [10, 20],  
    'dept_name': ['Engineering', 'HR']  
})
```